NAME : CHETHAN S
SRN : PES1UG25AM095
ROLL : 17
SECTION : P5

WEEK – 12 C PROGRAMS

CODE

```c
// PES1UG25AM095
#include<stdio.h>
#include<stdlib.h>

struct Node
{
    int data;
    struct Node* next;
};


void display(struct Node** head_reference)
{
struct Node* temp = *head_reference;
printf("The linked list is :  \n\n");
while(temp != NULL)
{
    printf("%d -> ", temp->data);
    temp = temp->next;
}
printf("NULL\n");
}

int main()
{
    struct Node *head = NULL  , *temp , *new_node;

    int n , i=0;

    printf("Enter number of nodes : ");
    scanf("%d",&n);

    while(i<n)
    {
        new_node = (struct Node *)malloc(sizeof(struct Node));
        printf("Enter data for node %d : ",i+1);
        scanf("%d",&new_node -> data);
        new_node -> next = NULL;

        if(head == NULL)
        {
            head = new_node;
            temp = new_node;
        }
        else
        {
            temp -> next = new_node;
            temp = temp -> next;
        }
        i++;
    }
```

PROGRAM 1

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK - 12> cd "c:\
Enter number of nodes : 5
Enter data for node 1 : 12
Enter data for node 2 : 23
Enter data for node 3 : 34
Enter data for node 4 : 45
Enter data for node 5 : 56
The linked list is :

12 -> 23 -> 34 -> 45 -> 56 -> NULL
PS C:\Users\Chethan S\Documents\WEEK - 12>
```

CODE

PROGRAM 2

```c
// PES1UG25AM095
#include<stdio.h>
#include<stdlib.h>

struct Node
{
    int data;
    struct Node* next;
};


void display(struct Node** head_reference)
{
struct Node* temp = *head_reference;
printf("The linked list is :  \n\n");

while(temp != NULL)
{
    printf("%d -> ", temp->data);
    temp = temp->next;
}
printf("NULL\n");

}

void insert_at_begining(struct Node** head_reference , int new_data)
{
    struct Node* new_node = (struct Node *)malloc(sizeof(struct Node));
    new_node -> data = new_data;
    new_node -> next = *head_reference;

    *head_reference = new_node;

}


void insert_at_end(struct Node** head_reference , int new_data)
{
    struct Node* new_node = (struct Node *)malloc(sizeof(struct Node));
    new_node -> data = new_data;
    new_node -> next = NULL;



    if(*head_reference == NULL)
    {
        *head_reference = new_node;
    }
    else
    {
        struct Node* temp = *head_reference;

        while(temp->next != NULL)
        {
            temp = temp->next;
        }
        temp->next = new_node;
    }
}

void delete_at_begening(struct Node** head_reference)
{
    if(*head_reference == NULL)
    {
        printf("There is no linked list to delete!!!");
        return;
    }

    struct Node* temp = *head_reference;

    (*head_reference) = (*head_reference) -> next;

    free(temp);
    printf("Deleted first node successfully!!!!");

}

void delete_at_end(struct Node** head_reference)
{
    if(*head_reference == NULL)
    {
        printf("There is no linked list to delete!!!");
        return;
    }

    struct Node* current = *head_reference;
    struct Node* previous = NULL;

    while(current -> next != NULL)
    {
        previous = current;
        current = current -> next;
    }
    previous -> next = NULL;

    printf("Deleted Last node successfully!!!!");

}
int main()
{
    struct Node *head = NULL  , *temp , *new_node;
    int new_data1 , new_data2;
    int n , i=0;

    printf("Enter number of nodes : ");
    scanf("%d",&n);

    while(i<n)
    {
        new_node = (struct Node *)malloc(sizeof(struct Node));
        printf("Enter data for node %d : ",i+1);
        scanf("%d",&new_node -> data);
        new_node -> next = NULL;

        if(head == NULL)
        {
            head = new_node;
            temp = new_node;
        }
        else
        {
            temp -> next = new_node;
            temp = temp -> next;
        }
        i++;
    }

    display(&head);

    printf("Enter the new data to be inserted at begining  : ");
    scanf("%d",&new_data1);

    insert_at_begining(&head , new_data1);
    display(&head);

    printf("Enter the new data to be inserted at end    : ");
    scanf("%d",&new_data2);

    insert_at_end(&head , new_data2);
    display(&head);

    delete_at_begening(&head);
    display(&head);

    delete_at_end(&head);
    display(&head);

    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK - 12> cd "c:\Users\Chethan S

23 -> 34 -> 45 -> 56 -> NULL
Enter the new data to be inserted at begining  : 12
The linked list is :

12 -> 23 -> 34 -> 45 -> 56 -> NULL
Enter the new data to be inserted at end    : 67
The linked list is :

12 -> 23 -> 34 -> 45 -> 56 -> 67 -> NULL
Deleted first node successfully!!!!The linked list is :

23 -> 34 -> 45 -> 56 -> 67 -> NULL
Deleted Last node successfully!!!!The linked list is :

23 -> 34 -> 45 -> 56 -> NULL
PS C:\Users\Chethan S\Documents\WEEK - 12>
```